

Homework problems:

Exercise 6.1 (15 points). Solve all the exercises shown on the page Problem H.6.1 - Monte Carlo until May 27, 09:00.

Remarks:

1. The points for this problem count for the grade bonus. Hence, you will have to upload your solution to Moodle until **May 27, 09:00**. Use the link given on Moodle under the topic "Peer-Review of Homework Problems" to upload your solution.

Your upload consists of the Quarto file containing your solution **and** the HTML file, that you generated by rendering the Quarto file.

Note that you **SHOULDN'T** write your name on your submission. You are identified by signing in on Moodle.

2. You can earn up to **five additional points**. To do this, you need to grade three solutions (your own and two from your peers) that will be assigned to you on Moodle for evaluation. You can gain up to five additional points based on the grading quality. See next week's problem set for more details.

3. Note that there is a template file `starter_mc.qmd` on Moodle for this problem (in the folder *Data, Quarto and R files - Sample solutions and previous problem sets*). You can certainly change the name of the file, but don't choose your name as file name.

In case you don't use the template file, make sure to include in the YAML header the option `embed-resources: true`:

```
format:
  html:
    embed-resources: true
```

Solution: Check again the page Problem H.6.1 - Monte Carlo to find the solution to the problem as well as some **grading instructions**.

Exercise 6.2. Let X and Y be random variables with $\text{Var}[X] = 3$, $\text{Var}(Y) = 3$, and $\text{Var}[3X+3Y] = 90$. Calculate the correlation $\text{Corr}[X, Y]$.

Solution:

According to Section 7.7.3, we have:

$$\text{Var}[X + Y] = \text{Var}[X] + \text{Var}[Y] + 2\text{Cov}[X, Y]$$

and it holds that:

$$\text{Var}[bX] = b^2\text{Var}[X] \quad \text{and} \quad \text{Cov}[bX, dY] = bd \cdot \text{Cov}[X, Y], \quad \forall b, d \in \mathbb{R}.$$

Thus, we obtain:

$$\begin{aligned}
\text{Var}[3X + 3Y] &= \text{Var}[3X] + \text{Var}[3Y] + 2\text{Cov}[3X, 3Y] \\
&= 9\text{Var}[X] + 9\text{Var}[Y] + 2 \cdot 3 \cdot 3 \cdot \text{Cov}[X, Y] \\
&= 27 + 27 + 18 \cdot \text{Cov}[X, Y] = 90
\end{aligned}$$

Therefore, the covariance is equal to

$$\text{Cov}[X, Y] = \frac{90 - 27 - 27}{18} = 2,$$

and this yields a correlation of

$$\text{Corr}[X, Y] = \frac{\text{Cov}[X, Y]}{\sqrt{\text{Var}[X] \cdot \text{Var}[Y]}} = \frac{2}{\sqrt{3 \cdot 3}} \approx 0.667.$$

Exercise 6.3. Let f be the function defined on $\mathbb{R}$ by

$$f(x) = ax^4 \cdot \mathbf{1}_{[0,1]}(x),$$

where a is a constant.

- a) Determine a such that f is a density function.
- b) The random variable X has density f with a computed in a). Calculate the expected value and the variance of X .

Solution:

- a) The function f is non-negative if $a \geq 0$. From

$$\int_{\mathbb{R}} f(x) dx = \int_0^1 ax^4 dx = a \left. \frac{x^5}{5} \right|_0^1 = a \cdot \frac{1}{5}$$

it follows that $a = 5$. Thus, $f(x) = 5x^4 \cdot \mathbf{1}_{[0,1]}(x)$ is the density of a continuous distribution, specifically the Beta distribution $\text{Beta}(5, 1)$.

- b) The expected value of $X \sim \text{Beta}(5, 1)$ is

$$\mathbb{E}[X] = \int_{\mathbb{R}} xf(x) dx = 5 \int_0^1 x^5 dx = 5 \cdot \left. \frac{x^6}{6} \right|_0^1 = \frac{5}{6}$$

and the variance is equal to

$$\begin{aligned}
\text{Var}[X] &= \mathbb{E}[X^2] - \mathbb{E}[X]^2 = 5 \cdot \int_0^1 x^6 dx - \frac{25}{36} = 5 \cdot \left. \frac{x^7}{7} \right|_0^1 - \frac{25}{36} \\
&= \frac{5}{7} - \frac{25}{36} = \frac{180 - 175}{252} = \frac{5}{252} \approx 0.0198.
\end{aligned}$$